CIS 678 Machine Learning

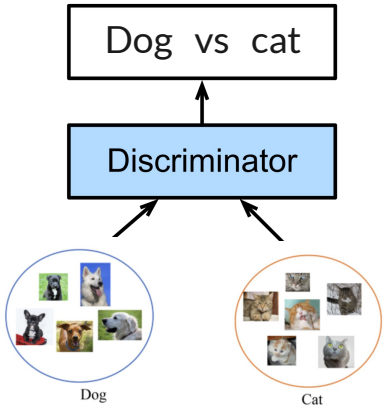
Special Topics (part 3)

Outline

Introduction to

- Recurrent Neural Networks (RNN)
- Transformers
- **Generative Modeling**
- **Foundational Models**

Generative AI

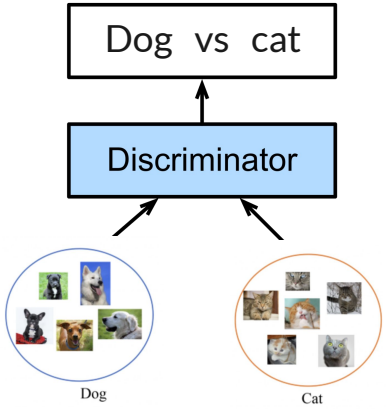


ref

Generative AI

Examples:

- Generative Adversarial Networks (GANs)



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Generative AI

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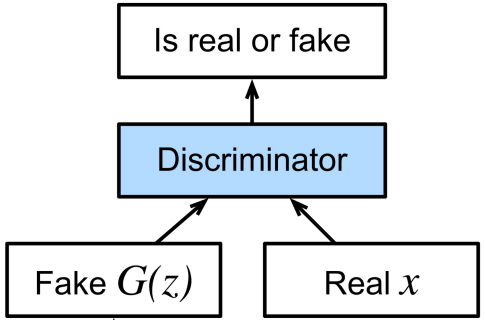

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Generative AI

Examples:

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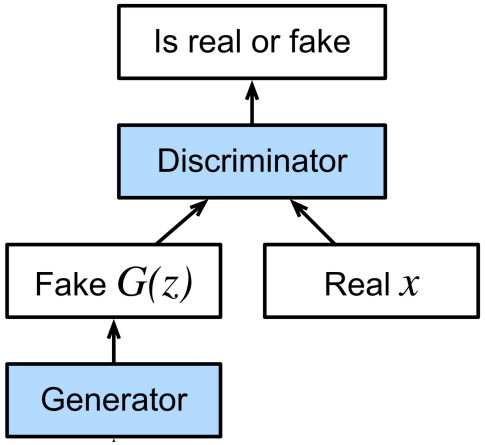

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Generative AI

Examples:

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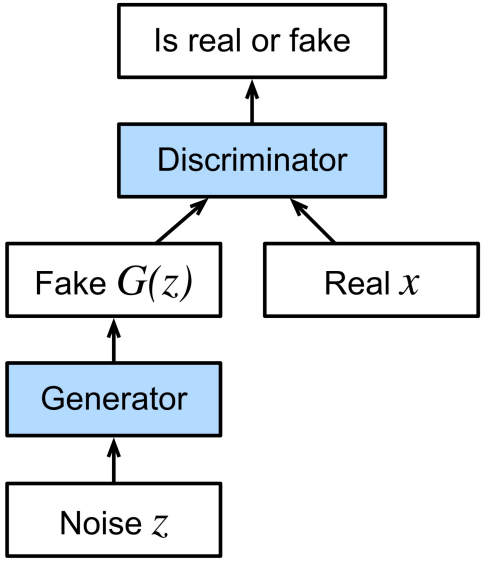

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Generative AI

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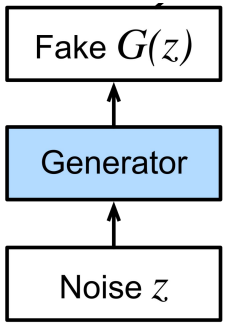

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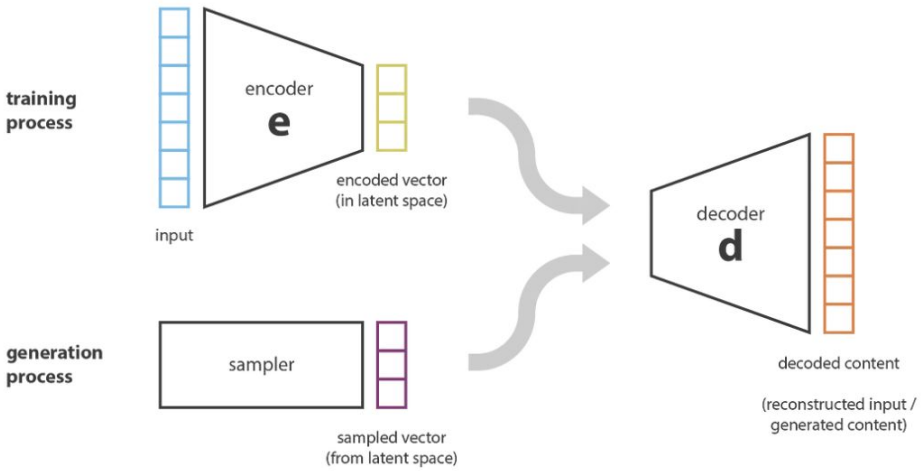
Generative AI

Examples:

- Generative Adversarial Networks (G
- **Variational Autoencoders (VAEs)**



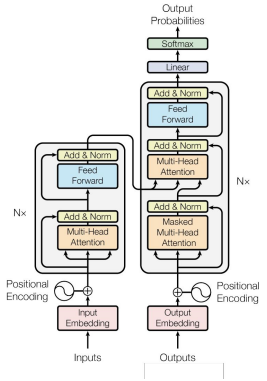
ref

Foundational models

- We don’t train independent models explicitly (say machine translation)

BERT

Encoder



Machine translation

GPT

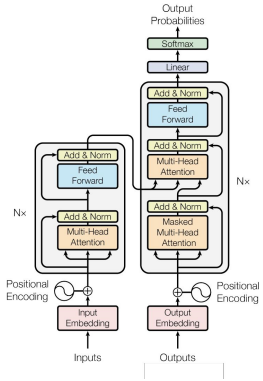
Decoder

Foundational models

- We don’t train independent models explicitly (say machine translation)
- Or document summarization

BERT

Encoder



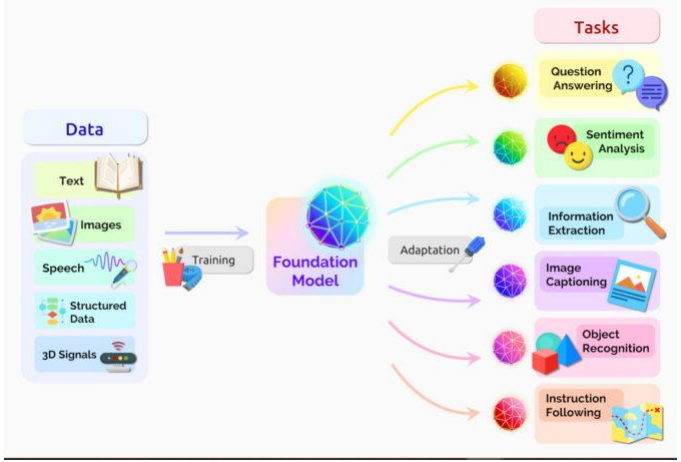
GPT

Decoder

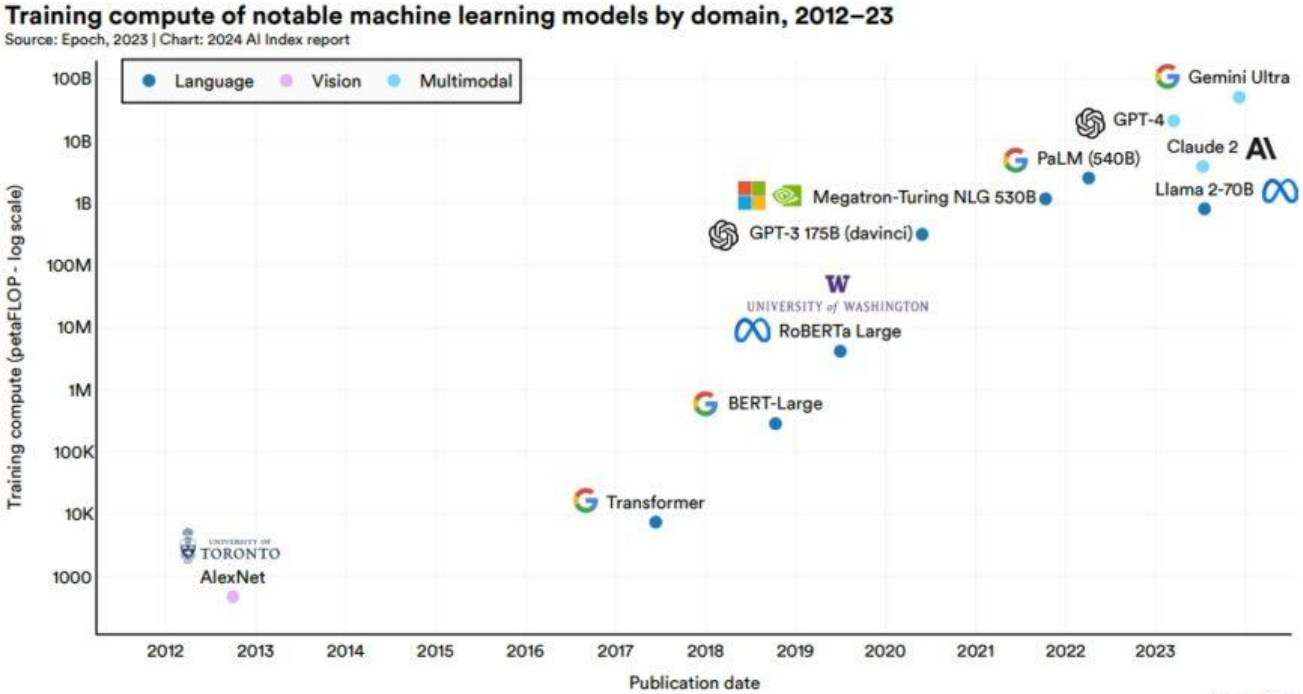
Document summarization

Foundational models

- We don’t train independent models explicitly (say machine translation)
- Or document summarization
- **We train a Base model for multiple tasks jointly, and then fine tune for specific tasks**



Foundational models



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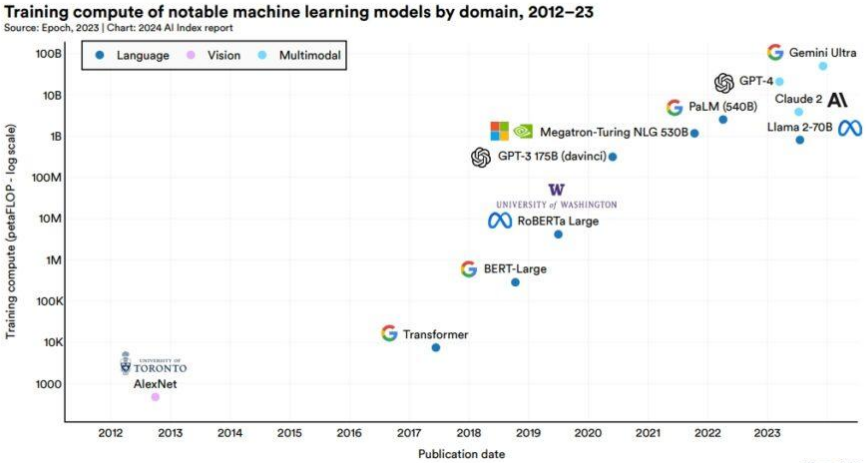

Figure 1.3.7

QA